MACHINE LEARNING FUNDAMENTALS

General Information

Applied Computer Science

Specialization Artificial Intelligence

Andy Louwyck

2025 – 2026

vives University of Applied Sciences

Lecturer

Andy Louwyck

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- Doctor & Master of Science
- Associate Degree in Informatics
- Microdegree in AI & Data Science

2001 - 2005	PhD Fellow Groundwater Modeling (UGent)
2006	Science Teacher (HH Ninove)
2007 - 2008	Project Engineer Integral Water Management (IMDC)
2009 - 2020	Groundwater Modeler & Data Scientist (VMM)
2020 - 2022	AI Researcher (Vives)
2020 - …	Lecturer AI (Vives)
2023 - …	AI Coordinator (VMM)

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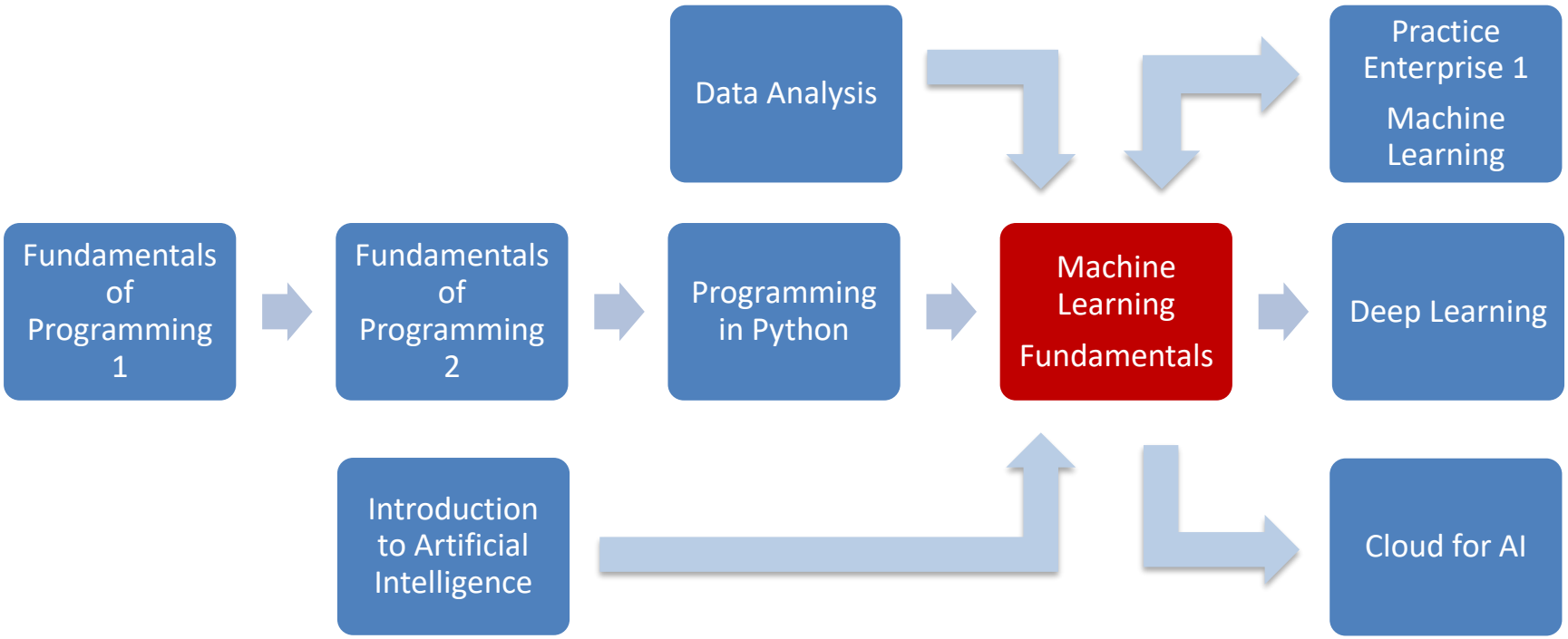
Positioning within the AI Curriculum



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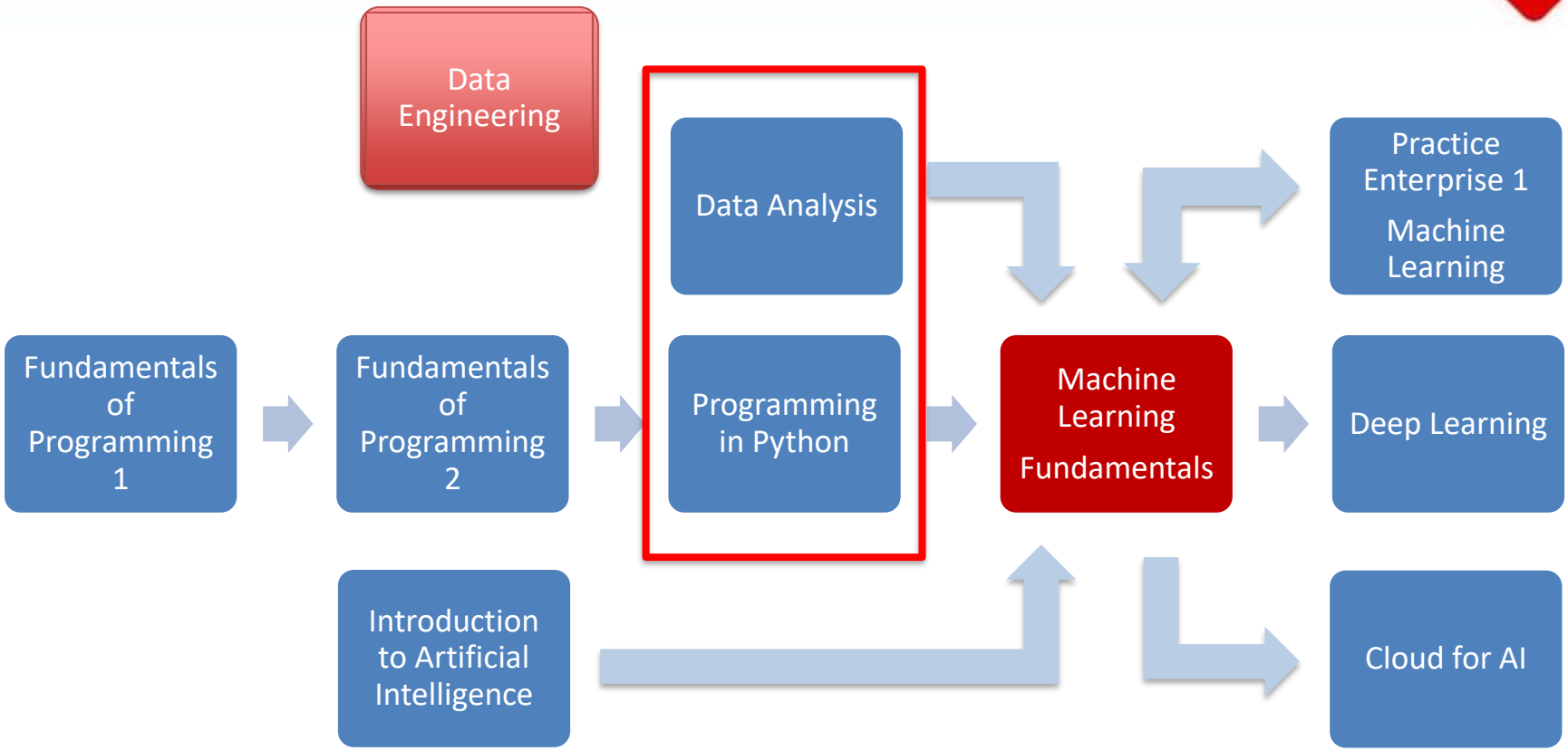
Positioning within the AI Curriculum



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Practical Arrangement & Evaluation

- **Contact:** 48 hours (6 ECTS) = 12 sessions of 4 hours
  - On Wednesday from 1 pm to 5:15 pm in room H60.11
  - Classes may be cancelled: see Toledo announcements!
  - Please notify me in advance if you will be absent!

- **Learning materials:** Toledo
  - Theory: slides + Jupyter notebooks + books + videos
  - Exercises: Jupyter notebooks (Google Colab)

- **Evaluation:** more info will follow later
  - Permanent: 20%
  - Exam: 80%

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Schedule (preliminary)

session	1	2			3	4	5	6	7			8	9	10	11
date	4 feb	11 feb	18 feb	25 feb	4 mar	11 mar	18 mar	25 mar	1 apr	8 apr	15 apr	22 apr	29 apr	6 may	13 may
topic	1	1	holidays	project week	2	3	4+5	6+7	8+9	holidays		10+11	12	13	14

1 intro
2 EDA
3 regression (linear + polynomial)
4 linear classification (perceptron + logistic regression)
5 support vector machines
6 evaluation metrics
7 k-nearest neighbors
8 decision trees
9 ensemble learning
10 clustering
11 principal component analysis
12 naive bayes
13 neural networks
14 timeseries

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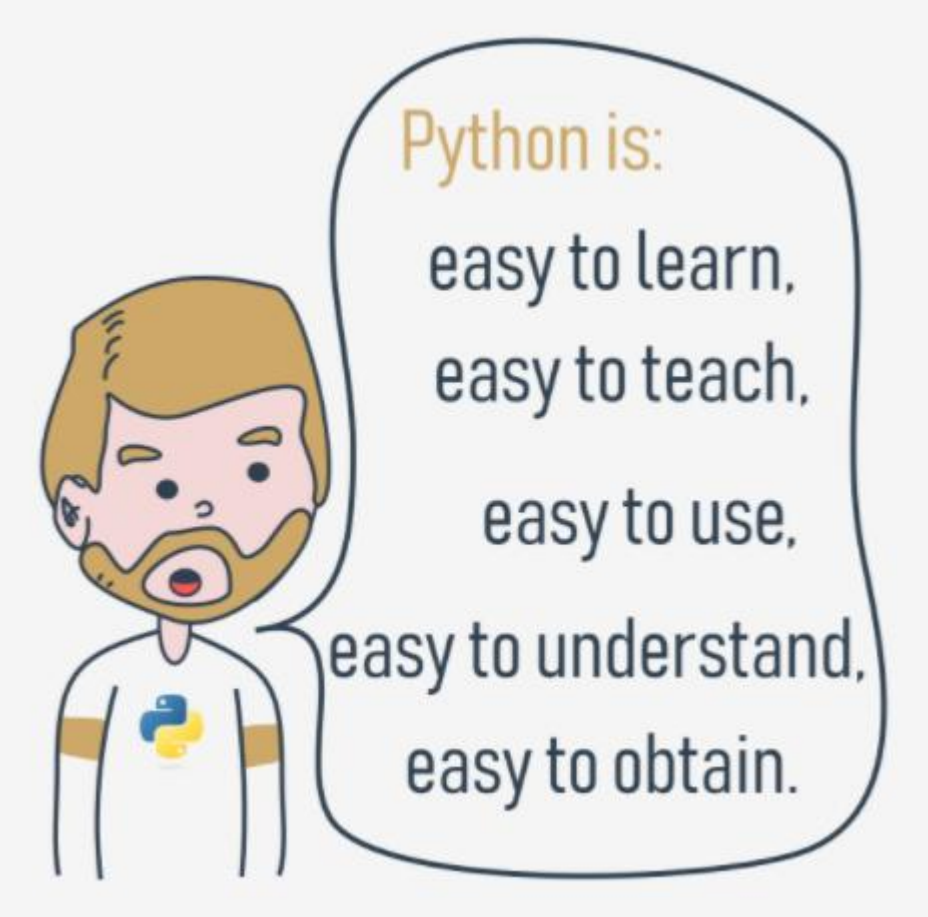
Why Python?


Python is:
easy to learn,
easy to teach,
easy to use,
easy to understand,
easy to obtain.

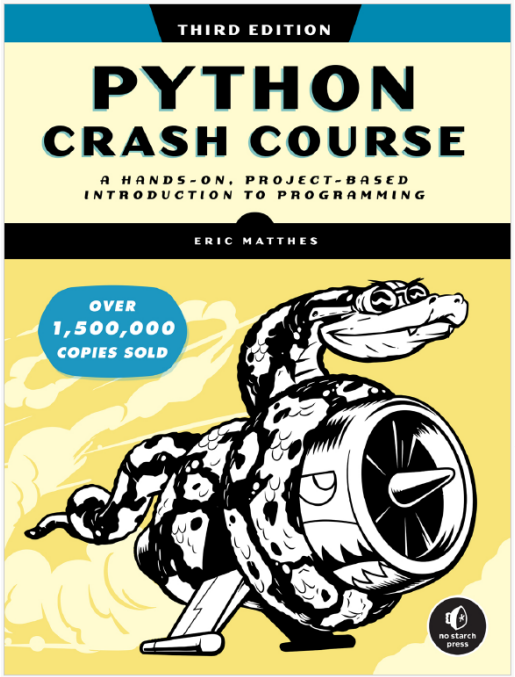
Python is global

Recommended Reading: Python Basics

- Python 101 notebooks
- VIVES online Python course
- Python tutorial on W3Schools
- Python Crash Course (3ed) by Eric Matthes

Python

- python_notebooks.zip
- Get Started with Google Colab - YouTube video
- VIVES online Python course
- Python tutorial on W3Schools
- Python Crash Course 3ed 2023 Eric Matthes.pdf



Popular Python Packages

seaborn

NumPy

theano

NLTK

Keras

plotly

mxnet

bokeh

SciPy

matplotlib

scikit learn

pandas

pillow

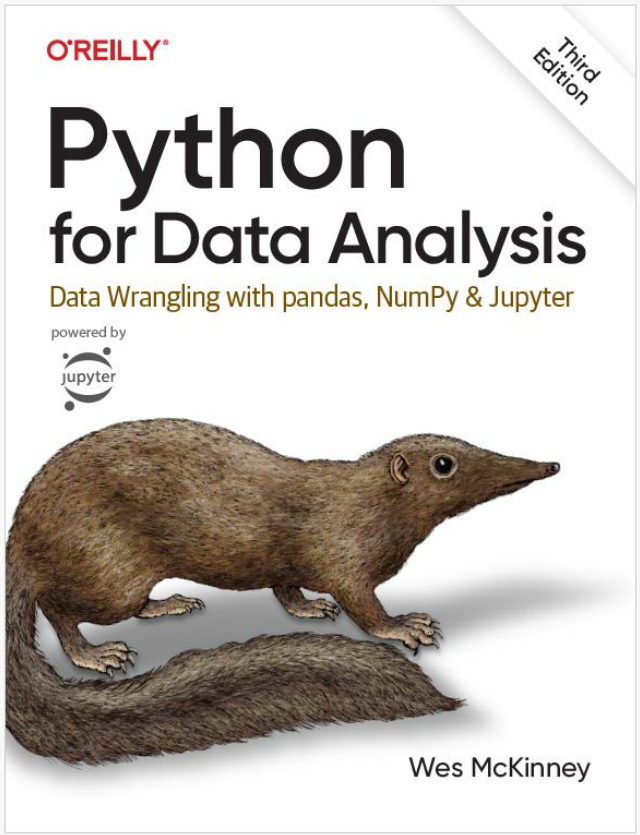
PYTORCH

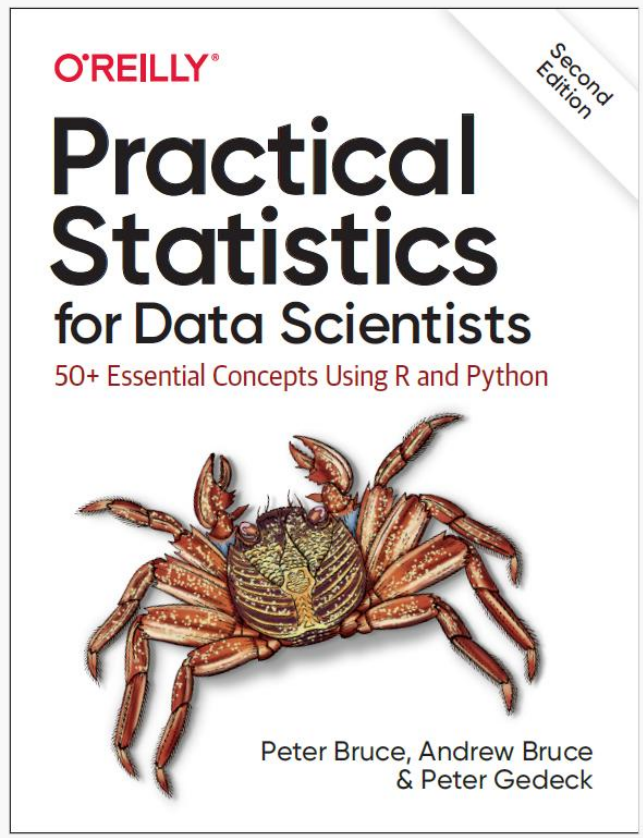
APACHE Spark MLlib

TensorFlow

Recommended Reading: Data Analysis

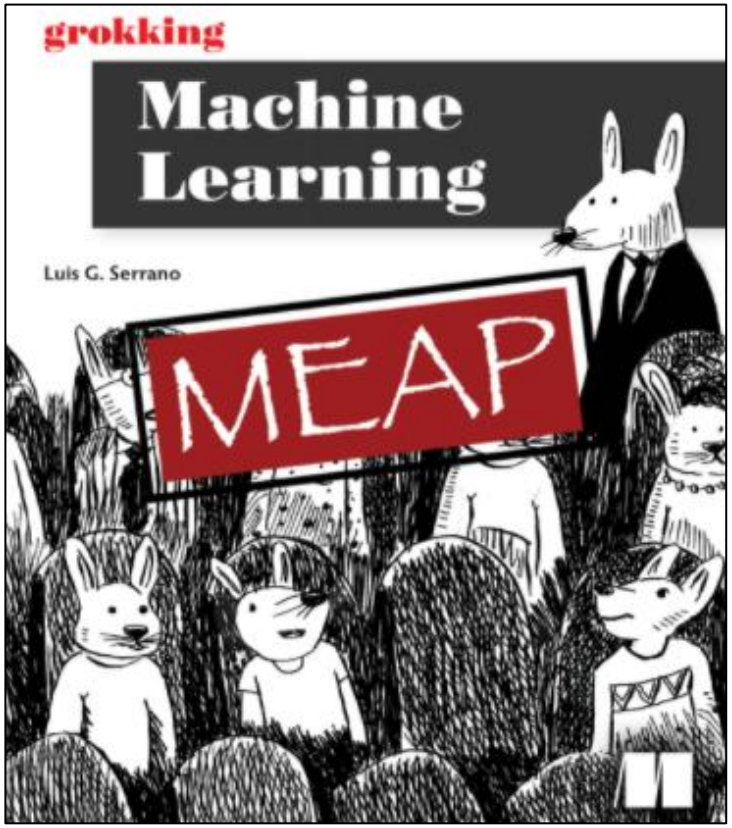
- Python for Data Analysis (3rd Ed) by Wes McKinney
- Practical Statistics for Data Scientists (2nd Ed) by Peter Bruce et al.

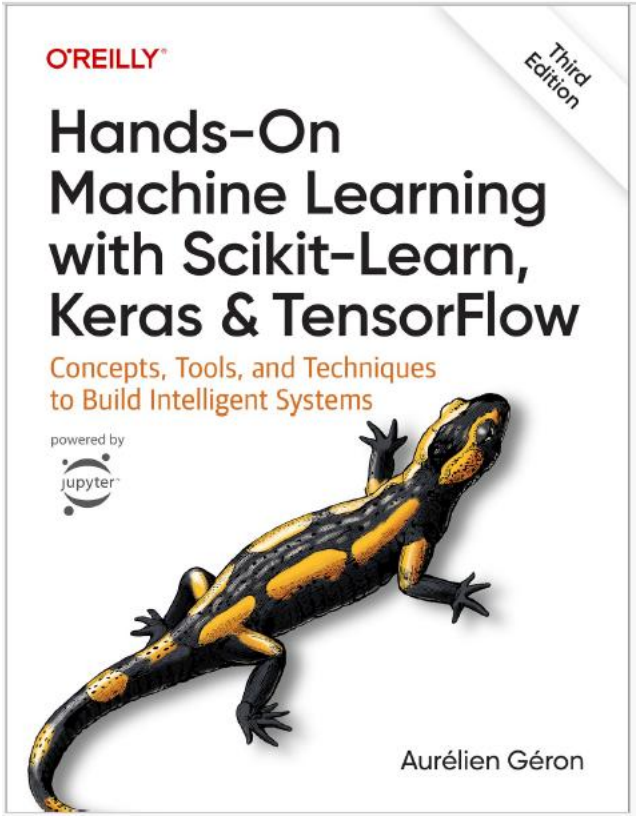




Recommended Reading: Machine Learning

- Grokking Machine Learning by Luis G. Serrano
- Hands-On Machine Learning (3rd Ed) by Aurélien Géron





How to ace this Machine Learning course?

- **Attend and engage actively during class**
  - Arrive on time and stay focused.
  - Avoid using laptops during lectures.

- **Study consistently throughout the semester**
  - Don't wait until just before the exam to study.
  - Focus on understanding, not rote memorization.
  - Explain concepts in your own words while studying.

- **Practice extensively**
  - Complete lots of exercises, beyond just class time.
  - Ask for help if stuck on theory or exercises.

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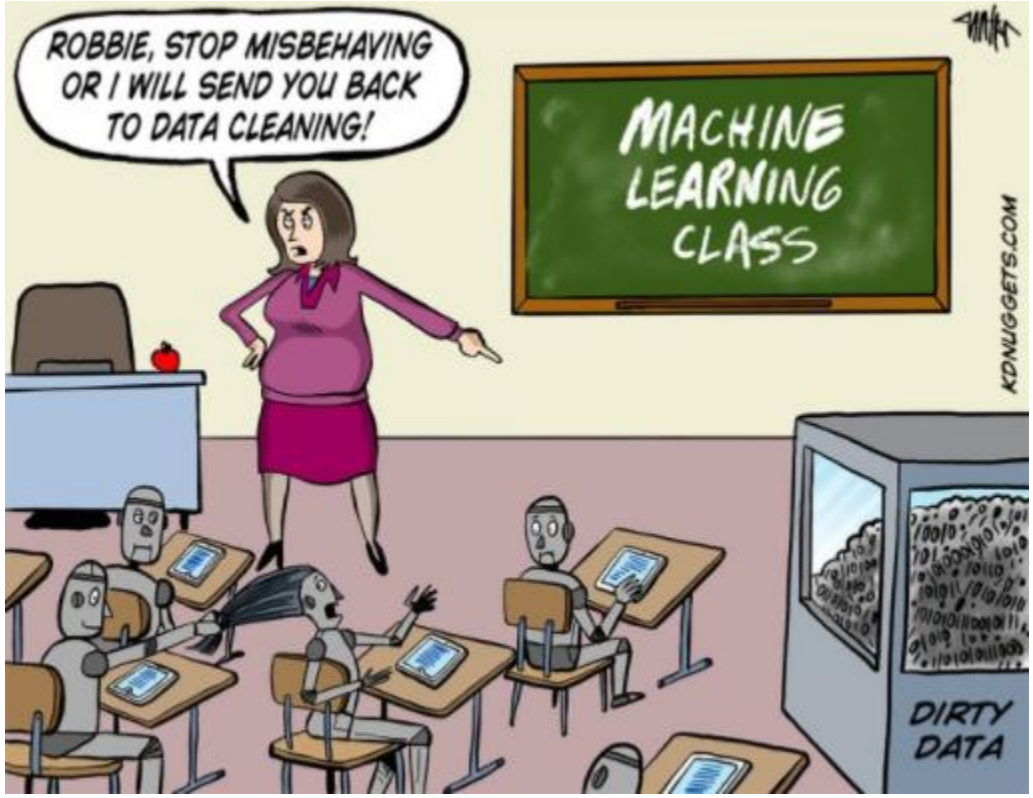
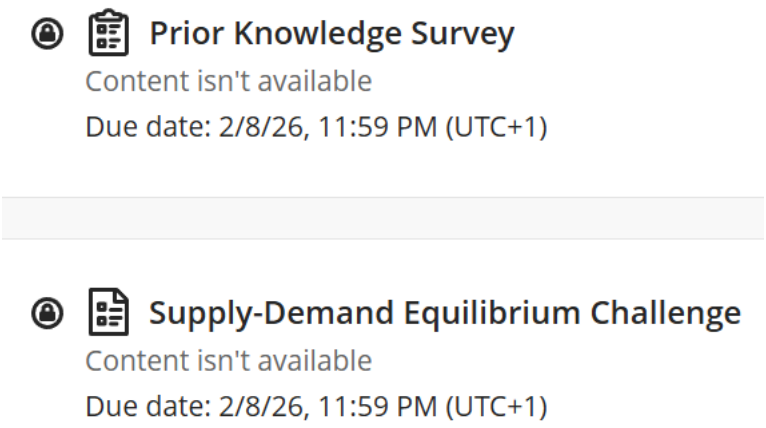
Have Fun!


ROBBIE, STOP MISBEHAVING OR I WILL SEND YOU BACK TO DATA CLEANING!
MACHINE LEARNING CLASS
KDNUGGETS.COM
DIRTY DATA

Challenge

- **See Toledo for instructions**
  - Explore the dataset
  - Develop a Python algorithm
  - Make the predictions
  - Present the results
- **Work in groups of 2, 3, or 4 students**
  - Choose a group name
  - Everyone submits the solution
  - Everyone fills in the survey
  - Deadline: **Sunday 8 February** at midnight



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